AapdaNetra: Disaster Intelligence Platform

“What-If?” Disaster Simulation Engine

Comprehensive Technical Specification, Algorithmic Mathematics & System Workflow

Document Version: 2.4.0 | **Classification:** Technical Reference | **Target Audience:** Disaster Commissioners, Incident Engineers
Core Service: simulationService.js | **Route:** POST /api/v1/intelligence/simulate | **UI:** frontend/src/pages/Simulation.jsx

1. Executive Overview & Purpose

The **“What-If?” Disaster Simulator** is an interactive predictive stress-testing sandbox in AapdaNetra. It empowers municipal commissioners, disaster response teams (NDRF/SDRF), and urban planners to hypothesize extreme climatological anomalies (e.g. cloudbursts, heatwaves, arid gale-force winds) and project their hydrodynamic and socio-economic aftermaths *before* catastrophe strikes. By combining real-time baseline telemetry, machine learning inference, GIS elevation profiles, and spatial demographics, the engine computes simulated risk escalations, displacement deltas, and municipal shelter deficits in sub-second response times.

System Architectural Flow

Pipeline Stage	Component	Key Responsibilities & Operational Logic
1. UI & Parameter Selection	Simulation.jsx React 18 + MUI	Captures geographic sector (lat/lon), scenario preset (5 types), and stress delta (+10% to +100%). Enforces strict simulation notice to prevent false public alerts.
2. Route & Validation	intelligenceRoutes.js Express + Joi/Custom	Routes payload via POST /api/v1/intelligence/simulate. Sanitizes latitude [-90, 90], longitude [-180, 180], and ensures percentage adjustment is bounded.
3. Baseline Telemetry	riskEngine.js Multi-Source Fusion	Pulls live meteorological conditions (OpenWeather), ML microservice predictions (Python FastAPI), citizen reports within 10km, and historical hazard polygons.
4. Perturbation Engine	simulationService.js Mathematical Modifiers	Applies scenario perturbation equations to baseline metrics. Calculates differential risk scores for FLOOD, LANDSLIDE, and WILDFIRE on a 0–100 scale.
5. Impact & Shelter Modeling	MongoDB Geospatial Habitation & Shelter	Filters habitations exceeding critical danger threshold (Score ≥ 50), tallies displaced citizens, computes active shelter capacity, and yields shelter bed deficit.

2. Environmental Perturbation & Scenario Formulation

The simulation engine supports five distinct disaster scenarios. When an operator selects a scenario and an intensity factor Δ (%) via the frontend slider, the engine applies non-linear meteorological transformations to the baseline parameters:

Scenario Code	Scenario Name & Focus	Mathematical Transformation
heavy_rainfall	Rainfall Inundation Surge (+10% to +100%)	$R_{sim} = R_{base} \times (1 + \Delta / 100)$ $H_{sim} = \min(H_{base} \times 1.15, 100\%)$
extreme_rainfall	Flash Downpour & Upstream Inflow (Cloudburst Simulation)	$R_{sim} = \max(R_{base}, 20mm) \times (1 + \Delta / 100)$ $H_{sim} = \min(95\%, H_{base} + 20\%)$ CloudCover = 95%
temperature_rise	Heatwave & Evaporation Surge (+1°C to +15°C)	$T_{sim} = T_{base} + \Delta°C$ $H_{sim} = \max(H_{base} - 2\Delta, 10\%)$
wildfire_conditions	Arid Wind Shift & Fuel Dryness (Fire Danger Index)	$T_{sim} = \max(T_{base}, 35°C) + 0.5\Delta$ $H_{sim} = \max(15\%, H_{base} - \Delta)$ $W_{sim} = W_{base} \times (1 + \Delta / 200)$ $R_{sim} = 0$ mm
landslide_rainfall	Sustained Slope Saturation (Geotechnical Pore Pressure)	$R_{sim} = \max(R_{base}, 30mm) \times (1 + \Delta / 100)$ $H_{sim} = \min(95\%, H_{base} + 15\%)$

3. Hydrodynamic & Risk Modification Equations

Once perturbed environmental values (R_{sim}, T_{sim}, H_{sim}, W_{sim}) are generated, the engine executes disaster-specific sensitivity functions to compute risk deltas:

Disaster Vector	Sensitivity & Delta Modifier Equation	Primary Physical Drivers
FLOOD RISK	**Modifier** $= (R_{sim} - R_{base}) \times 1.5 + (H_{sim} - H_{base}) \times 0.3$ $\mathbf{Score}_{\mathbf{new}} = \text{clamp}(Score_{base} + \text{Modifier}, 0, 100)$	Runoff volume, soil saturation, drainage capacity saturation.
LANDSLIDE RISK	**Modifier** $= (R_{sim} - R_{base}) \times 1.2 + (H_{sim} - H_{base}) \times 0.4$ $\mathbf{Score}_{\mathbf{new}} = \text{clamp}(Score_{base} + \text{Modifier}, 0, 100)$	Soil pore-water pressure, shear stress on slope toe.
WILDFIRE RISK	**Modifier** $= (T_{sim} - T_{base}) \times 2.0 + (H_{base} - H_{sim}) \times 0.8 + (W_{sim} - W_{base}) \times 1.5$ $\mathbf{Score}_{\mathbf{new}} = \text{clamp}(Score_{base} + \text{Modifier}, 0, 100)$	Vegetation desiccating temperature, relative humidity drop, wind propagation velocity.

Risk Classification Thresholds:

- **CRITICAL (Score ≥ 76):** Severe life hazard. Mandatory evacuation protocols and emergency warning.
- **RED (51 ≤ Score ≤ 75):** High danger. Immediate relief mobilization, shelter preparation.
- **AMBER (26 ≤ Score ≤ 50):** Elevated watch. Monitor sluice gates, standby emergency personnel.
- **GREEN (Score ≤ 25):** Normal baseline conditions. Low immediate threat.

4. Baseline Unified Risk Fusion Mechanism

Before applying any hypothetical adjustment, the simulator grounds itself in empirical reality by pulling the unified risk composite from riskEngine.js. The baseline composite score (0–100) is determined by a weighted multi-layer formula:

Component	Weight	Data Source & Methodology
Machine Learning Prediction	**40%**	FastAPI Python microservice running Random Forest & XGBoost models on live weather features (temp, humidity, rain, wind, pressure, soil moisture).
Real-time Weather Risk	**25%**	Live OpenWeather API readings mapped to hazard sensitivity curves.
Historical Spatial Hazards	**15%**	Geo-spatial query within 15km for historical floodplains and seismic zones in MongoDB HazardZone collection.
Habitation Vulnerability	**10%**	Average structural vulnerability index of habitations within 10km radius.
Crowdsourced Citizen Reports	**10%**	Verified citizen reports in the last 24h within 10km (each report adds 5 pts up to 20 max boost).

5. Municipal Impact & Shelter Deficit Modeling

A critical innovation of AapdaNetra's simulator is translating abstract risk numbers into actionable municipal logistics:

1. Affected Habitations Thresholding:
The system scans all habitations in the jurisdiction. Any habitation where:
$(\text{CurrentRiskScore} + \text{MaxHazardChange}) \geq 50$
is flagged as an endangered settlement requiring immediate operational intervention.

2. Displaced Population Tally:
Aggregates total census headcount: $P_{affected} = \sum \text{Population}(h_i)$ across all endangered settlements.

3. Live Shelter Availability:
Queries all operational emergency shelters (status ≠ 'CLOSED'):
$\text{Capacity}_{avail} = \sum (\text{MaxCapacity}_j - \text{CurrentOccupancy}_j)$

4. Projected Shelter Deficit & Auxiliary Relief Tents:
$\text{Deficit} = \max(0, P_{affected} - \text{Capacity}_{avail})$
This metric immediately informs district commissioners exactly how many auxiliary relief tents, mobile sanitation units, and food parcels must be requisitioned.

6. Fail-Safe Sandbox Isolation & Life-Safety Guardrails

Because false disaster alarms cause panic, civil disruption, and alert fatigue, AapdaNetra enforces strict architectural isolation:

• **Zero Sound Output:** The Civil Defense Siren (playEmergencySiren()) and browser Web Audio synthesizers are completely disarmed inside the Simulation sandbox.
• **No Citizen Push Dispatches:** Simulation runs never emit push notifications, WebPush events, or WhatsApp/SMS alerts to residents.
• **Database Read-Only Isolation:** Simulation calculations run purely in-memory. No synthetic risk scores are persisted into production Alert or CitizenReport tables.
• **Mandatory Simulation Watermark:** All output payloads carry a mandatory disclaimer header: '■■ *SIMULATION — This is a hypothetical scenario, NOT a real prediction or forecast.'*

7. End-to-End Walkthrough: Delhi Yamuna +50% Surge Case Study

Consider an emergency planner in Delhi testing a severe monsoon cloudburst over the Yamuna Floodplain sector:

Simulation Parameter	Baseline Value	Simulated Value	Net Impact / Delta
Target Micro-Sector	Yamuna Floodplain R-12	Yamuna Floodplain R-12	Sector Coordinates: 28.6139°N, 77.2090°E
Monsoon Precipitation	18.4 mm	**27.6 mm** (+50%)	+9.2 mm surge in 2-hour window
Relative Humidity	74%	**85.1%**	+11.1% near-saturation atmospheric vapor
Flood Threat Score	48 / 100 (AMBER)	**65 / 100 (RED)**	**+17 pts** escalation into high hazard
Endangered Habitations	2 settlements	**7 settlements**	+5 vulnerable settlements submerged
Affected Resident Population	4,200 citizens	**22,100 citizens**	+17,900 citizens requiring immediate evacuation
Available Shelter Beds	1,965 beds	1,965 beds	Existing static municipal shelters at full capacity
Projected Shelter Deficit	2,235 beds	**20,135 beds**	**Urgent requisition of 40+ auxiliary relief camps required**

Conclusion & Tactical Value:
By simulating this scenario in advance, the district magistrate does not need to guess shelter requirements during a midnight flood. Within 2 seconds, the simulator provides exact deficit figures (20,135 beds), identifies priority low-lying settlements (Yamuna Vihar, Nala Colony), and models logistics before rainfall even commences.